



Review of ophthalmic emergencies in primary care: a comprehensive approach to red eye

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Abstract: Red eye affects patients of all ages and is one of the most common ocular complaints seen in primary care and emergency rooms. Some conditions, like scleritis, keratitis, uveitis, or endophthalmitis, can be vision-threatening and necessitate an immediate diagnosis and referral, but many others, like conjunctivitis or episcleritis, are benign and self-limiting. To prevent complications, such as irreversible vision loss or the spread of infection throughout the body, it is essential to accurately and promptly distinguish between benign and serious causes. With a broad range of etiologies covered, this review offers a thorough, evidence-based method for assessing red eye in primary care. For inflammatory, infectious, traumatic, and non-inflammatory disorders, we look at important clinical characteristics, diagnostic factors, and therapeutic approaches. Dry eye syndrome, pterygium, scleritis, episcleritis, keratitis, conjunctivitis, uveitis, blepharitis, hordeolum, endophthalmitis, orbital cellulitis, hyphema, trichiasis, and subconjunctival hemorrhage are among them. Whereas episcleritis usually responds to topical treatment, inflammatory conditions like scleritis and uveitis need systemic corticosteroids or immunosuppressive therapy. Infections such as keratitis, conjunctivitis, and endophthalmitis require treatments specific to the pathogen, while diseases like orbital cellulitis and hyphema may require hospitalization or surgery. External conditions like dry eye and blepharitis are treated conservatively with lubrication and cleanliness. This review aims to provide general practitioners with useful clinical tools for timely assessment, efficient triage, and referral by defining red flag symptoms, hallmark signs, and updated treatment protocols. Regular eye exams and patient education also improve results and lower recurrence.

Keywords: Red eye; ocular emergencies; primary care; conjunctivitis; ocular inflammation

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Introduction

Up to 3 percent of visits to primary care doctors are related to eye issues (1). Red eye is among the most common presenting complaints in both primary and specialized healthcare (2). While many cases resolve spontaneously, some indicate serious or sight-threatening conditions that require prompt recognition. Early detection and treatment are essential for a favorable prognosis, supported by regular eye examinations and patient education. Red eye accounts for about 15% of ophthalmology consultations and 6% of visits to general practitioners in Eastern Europe (3). Causes range from mild to severe, making diagnosis challenging for non-ophthalmologists (4). Common causes include conjunctivitis, scleritis, episcleritis, uveitis, acute angle-closure glaucoma (AACG), and foreign bodies. Pain and/or photophobia help distinguish benign from serious conditions such as AACG, bacterial keratitis (BK), scleritis, or anterior uveitis (5). Recognizing clinical indicators like pupil changes, photophobia, and anterior chamber findings is necessary for differentiation (*Table 1*).

Rationale and knowledge gap

Despite the high prevalence of red eye in primary care and emergency settings, many frontline clinicians lack the specialized training to accurately differentiate benign cases from those requiring urgent ophthalmologic referral. Misdiagnosis or delayed recognition of vision-threatening conditions can lead to severe complications, including irreversible vision loss. There is a need for a structured, evidence-based approach to assist general practitioners and emergency physicians in efficiently evaluating and managing red eye presentations.

Objective

The objective of this review is to provide a comprehensive, clinically oriented framework for non-ophthalmologists to systematically assess and manage patients presenting with red eye (*Figure 1*). By consolidating updated diagnostic strategies and management options across a wide range of etiologies, this review aims to support timely recognition of vision-threatening conditions, guide appropriate referral decisions, and ultimately improve patient outcomes in primary care and emergency settings.

Infection and inflammation

Scleritis

Scleritis is a chronic inflammatory condition of the sclera that can be linked to systemic diseases or infections (6). It is more prevalent in females than males and typically affects middle-aged individuals (7). This condition may present with a gradual or acute onset, severe radiating pain exacerbated by ocular movement or touch, along with photophobia, tearing, and/or decreased visual acuity. Rarer subtypes of scleritis, such as scleromalacia perforans, demonstrate minimal symptomatology. A slit-lamp examination can evaluate for fixed, injected, and inflamed blood vessels and the extent of involvement (8). Providers should also assess for a violaceous hue to the sclera in natural light and a history of autoimmune symptoms such as rash or arthritis, which may point towards severe systemic illness (9). Scleritis is divided into anterior, which includes diffuse, nodular, and necrotizing subtypes, and posterior, which also features diffuse and nodular forms (10). The sclera's anatomical structure comprises an extracellular matrix of collagen, elastin, and proteoglycans and the exact pathophysiology of scleritis remains a focus of ongoing research (10). Rao *et al.* examined tissue specimens from patients with end-stage scleritis and identified two histological patterns: rheumatoid arthritis-associated scleritis showed tissue necrosis with surrounding granulomatous inflammation, while idiopathic scleritis exhibited chronic inflammatory infiltrate without necrosis (11). Usui *et al.* (2008) further analyzed the inflammatory cells in necrotizing scleritis tissues using immunohistochemical staining, finding a dominance of B cells (CD20⁺) and macrophages (CD68⁺) in autoimmune-related cases, and macrophages followed by T cells (CD3⁺) in idiopathic cases (12). The diagnosis of scleritis is primarily based on clinical presentation and ocular examination (*Table 1*). However, several laboratory tests can help rule out coexisting diseases. These include antineutrophil cytoplasmic antibodies (ANCA), antinuclear antibodies (ANA), HLA-B27, fluorescent treponemal antibody absorption (FTA-ABS) test, and C-reactive protein (CRP). First-line treatments for scleritis include topical corticosteroid eye drops, such as prednisolone acetate or difluprednate, administered four times daily. Oral nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen 600 mg or naproxen 500 mg, may also be added

Table 1 Clinical features of red eye conditions

Items	Scleritis	Episcleritis	Keratitis	Conjunctivitis	Uveitis	AACG
Pupil	Normal	Normal	Normal	Normal	Constricted or irregular	Mid-fixed, non-reactive
Pain	Severe	Mild	Moderate to severe	None or minimum	Moderate to severe	Moderate to severe
Photophobia	May be present	None	Present	None	Present consensual	Present
Anterior chamber	Clear	Clear	Exist cell and flare	Clear	Exist cell and flare	Shallow
Fluorescein	Uptake	No uptake	Uptake	No uptake	May show dendrites	No uptake

AACG, acute angle-closure glaucoma.

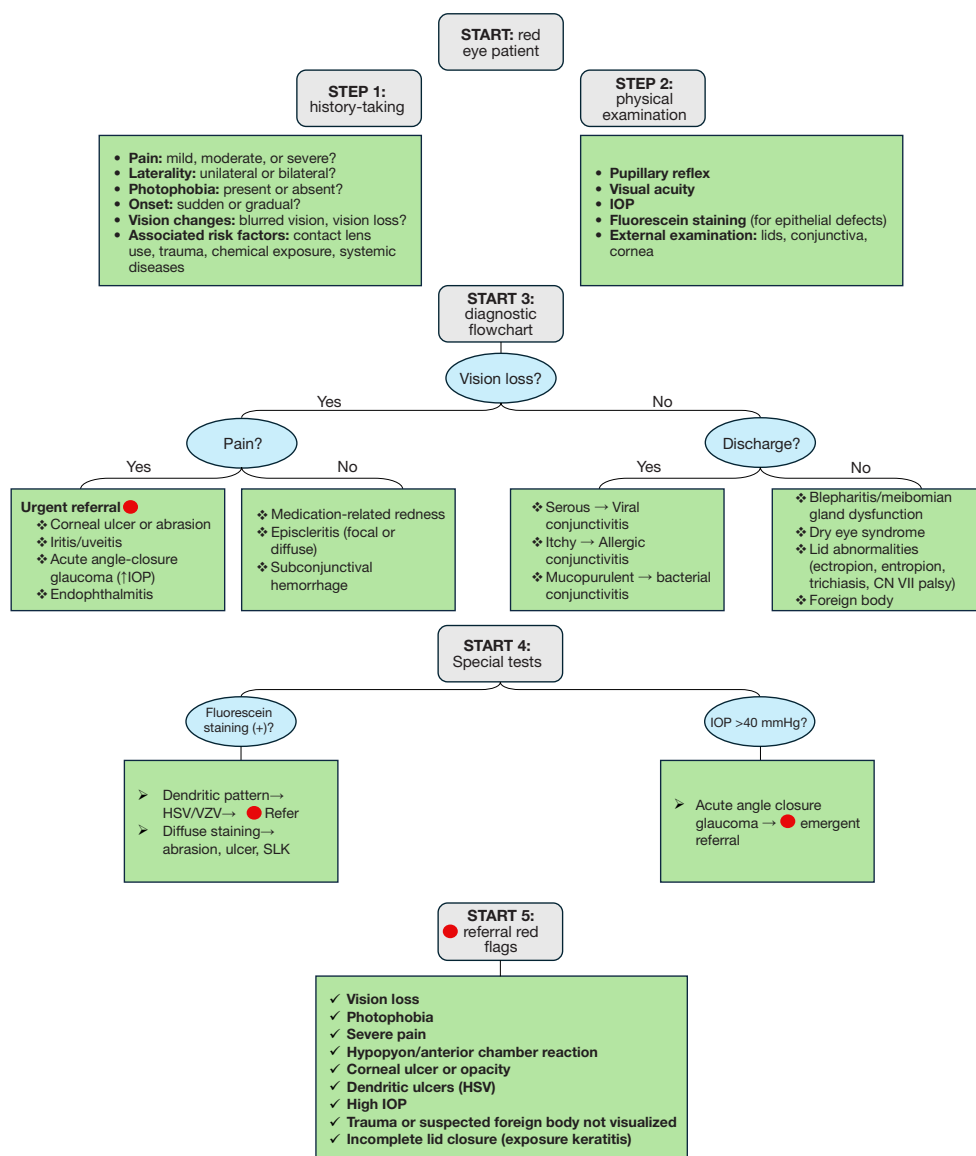


Figure 1 Algorithm for the diagnosis of red eye. This flowchart provides a structured approach to diagnosing red eye, guiding clinicians through a stepwise evaluation based on clinical findings. The red circles indicate findings that require urgent referral. CN, cranial nerve; HSV, herpes simplex virus; IOP, intraocular pressure; SLK, superior limbic keratoconjunctivitis; VZV, varicella zoster virus.

to the treatment regimen if needed. Severe cases may require oral corticosteroids (e.g., prednisone at 1 mg/kg) or immunosuppressive agents like methotrexate or biologic agents such as infliximab, especially in conjunction with systemic corticosteroids (13). Patients without previously discovered systemic diseases should undergo further workup, as up to half of scleritis cases are associated with underlying conditions, like connective tissue disorders or vasculitis (8).

Episcleritis

Episcleritis is inflammation of superficial vessels in the episclera, between the conjunctiva and the sclera (14). It presents with sudden sectoral or diffuse redness, usually without pain, vision changes, or discharge, though tenderness may occur. Incidence is <1/1,000, more common in women aged 40–50 (15). Diffuse episcleritis typically resolves within three weeks but may recur, while nodular episcleritis causes longer and more painful episodes (16). It is usually unilateral but can be bilateral or alternating. Most cases are idiopathic, though up to one-third are linked to systemic autoimmune disease (17). Mild episcleritis, often associated with dry eye, may be conservatively managed with supportive measures of cool compresses and artificial tears four times daily (8). Treatment for moderate to severe episcleritis generally involves the use of oral NSAIDs as a first-line therapy, or mild topical corticosteroids (18). However, it is important to note that steroid use may lead to recurrence of episcleritis. Patients receiving this treatment should be monitored closely every 2–3 weeks until full resolution of symptoms, followed by a gradual steroid taper (8).

Keratitis

Keratitis is corneal inflammation with edema, inflammatory cell infiltration, and ciliary congestion. Causes may be infectious or non-infectious, systemic or localized. Microbial keratitis is the most common and a major concern in developing countries (19), caused by bacteria, fungi, viruses, or parasites (20). While the exact prevalence is unclear, corneal ulceration occurs at about 11 per 100,000 annually in the USA (21) and affects males more often (22). Diagnosis relies on slit-lamp exam with fluorescein to assess infiltrates and epithelial damage; corneal scrapings are recommended for infiltrates >1–2 mm to identify pathogens by staining or culture (8). Infectious keratitis may follow ocular procedures such as small incision lenticule extraction

(SMILE), where the intrastromal junction favors pathogen invasion (23). These postoperative cases are difficult and require urgent management to prevent vision loss.

BK

BK is the most common microbial keratitis (24), presenting with painful epithelial and stromal ulceration, corneal opacity, melt, exudate, or hypopyon. Severe cases may progress to descemetocoele, perforation, and blindness (25). Major pathogens include *S. aureus*, *S. epidermidis*, *S. pneumoniae*, and *P. aeruginosa* (26), which typically infect only when corneal barriers are disrupted by factors such as contact lens use, corticosteroids, ocular surgery, diabetes, or connective tissue disease (27). Topical antibiotics remain first-line therapy (28). Moxifloxacin and gatifloxacin are widely used off-label (29). Kowalski *et al.* reported moxifloxacin had comparable empirical coverage (92%) to cefazolin/tobramycin (93%) and cefuroxime/gentamicin (97%), supporting its use as monotherapy while awaiting culture results (30). Vancomycin remains essential for methicillin-resistant *Staphylococcus aureus* (MRSA), highlighting the need for early susceptibility testing given rising resistance (31). Corneal melt involves fibroblast and keratocyte activity; tetracyclines may help by inhibiting collagenase (32). The role of topical corticosteroids is debated (33). Some suggest benefit in reducing stromal melt, neovascularization, and scarring (34), while others warn of delayed healing and worsening infection (35). Urgent ophthalmology referrals are crucial to prevent vision loss.

Viral keratitis

Viral keratitis is another common form of infectious keratitis, frequently caused by herpes simplex virus (HSV), adenovirus, varicella zoster virus (VZV), Epstein-Barr virus (EBV), and cytomegalovirus (CMV) (36). Viral keratitis can become chronic and recurrent, posing a painful and sight-threatening challenge to clinicians. Among these, the herpes group of viruses dominates as the most prevalent cause of viral keratitis worldwide. A characteristic dendritic ulcer, visible under fluorescein slit lamp exam, typically aids in the clinical diagnosis of herpes simplex keratitis (HSK), the most common form of viral keratitis (37). Additional clinical signs of HSK include stromal opacity and neovascularization, often triggered by recurrent HSV-1 reactivation and subsequent viral shedding into deeper layers of the cornea (38). HSK is classified into four subtypes based on the affected corneal layer and whether

the keratitis is due to primary or reactivated infection: (I) epithelial keratitis; (II) immune stromal keratitis; (III) stromal necrotic keratitis; and (IV) endotheliitis, with epithelial keratitis being the most common subtype (39). The virus's direct effects include blood vessel ingrowth, leukocyte infiltration, and damage to the corneal stroma and endothelium. Notably, HSV keratitis has been shown to significantly impact quality of life (40), with repeated episodes resulting in corneal scarring and potentially blindness. Topical antiviral medication (acyclovir) is the primary treatment for epithelial HSK (41). Topical steroids are the mainstay of treatment for HSV stromal disease and endotheliitis and are used in conjunction with antiviral medication to prevent further recurrence of viral keratitis (42). In cases of recurrent HSV keratitis, a prophylactic course of oral antiviral medication (acyclovir) is administered for one year (43).

Fungal keratitis (FK)

FK is a severe infection of the cornea marked by complications such as corneal scarring, perforation, stromal destruction, and ulceration (44). Corneal ulcers are most commonly caused by the major fungal pathogens, molds (*Fusarium* and *Aspergillus*), and yeasts (*Candida*) (45). Geographically, filamentous fungi are more prevalent in tropical and subtropical regions, whereas yeast is more prevalent in temperate areas (46). FK occurs when fungal spores or hyphal fragments penetrate the protective epithelium and reach the stroma (47). Corneal infiltrates in FK typically appear dry with feathery edges and multiple satellite lesions (48). Clinical features of FK include mild pain, reflecting the slow progression of fungal pathogens deep into the cornea (49). Topical Natamycin is the preferred treatment for filamentous FK (50), with topical voriconazole added for *Aspergillus* keratitis that does not respond to natamycin alone. Despite natamycin being first-line, its effectiveness can be limited by emerging drug resistance among fungal species (51).

Conjunctivitis

Conjunctivitis, commonly known as pink eye, is the inflammation of the conjunctiva, the lubricating mucous membrane covering the eye's outer surface. The conjunctiva is divided into two parts: the tarsal conjunctiva, which lines the inner eyelids, and the bulbar conjunctiva, which covers the eyeball up to the cornea (52). This condition affects millions of individuals annually and poses significant economic and social burdens (53). Conjunctivitis, or

inflammation of the conjunctiva, can result from allergic or immunological reactions as well as infections, with viruses and bacteria being the most common infectious causes (54). Ocular allergy is the most prevalent form of non-infectious conjunctivitis and can significantly impact quality of life (55). Diagnosis is typically clinical; however, Gram stains, cultures, or polymerase chain reaction (PCR) can aid in cases of atypical conjunctivitis, neonatal conjunctivitis, or undetermined etiologies. While most cases are self-limiting and rarely lead to vision loss, conjunctivitis remains a frequent concern amongst patients seeking ophthalmic care due to its uncomfortable symptoms and potential for widespread transmission (56).

Bacterial conjunctivitis (BC)

BC is more common in children, usually caused by *S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*; meningococcal cases may also occur (57). In adults, *Staphylococcus* predominates, while *N. gonorrhoeae* is a rare cause (57). BC arises from direct contact or overgrowth of native flora (56). It typically begins unilaterally with conjunctival hyperemia, foreign body sensation, and increased secretions (58), often leading to sticky eyelids and pruritus (59). Serious complications are uncommon (60), but keratitis risk increases in patients with corneal epithelial defects or dry eye. Contact lens wear should be avoided, as lenses may aggravate infection and harbor bacteria (61). Severe cases are best treated with the latest-generation fluoroquinolones; alternatives include polymyxin B/trimethoprim, ciprofloxacin, azithromycin solutions, or erythromycin and bacitracin ointments. In pediatrics, ciprofloxacin, ofloxacin, and tetracycline show the highest bactericidal activity, followed by gentamicin, tobramycin, and polymyxin combinations (62). Gonococcal conjunctivitis carries a high risk of perforation and blindness and requires immediate systemic ceftriaxone injection (63).

Viral conjunctivitis

Viral conjunctivitis is highly contagious and the most common form of conjunctivitis (64). It can be triggered by DNA viruses [e.g., adenovirus, HSV, and VZV], or RNA viruses (e.g., coxsackievirus, enterovirus, and picornavirus) (65). Among these, adenovirus is the most common cause (54). Infected individuals typically experience symptoms like watery discharge, ocular pruritus, tearing, and tender preauricular lymphadenopathy (66). Viral conjunctivitis is generally self-limiting and resolves within 2–4 weeks. In patients with current vesicular skin lesions or a history

of perioral fever blisters, a fluorescein examination is recommended, as HSV can cause corneal dendritic lesions (67). The standard treatment for viral conjunctivitis caused by adenoviruses includes artificial tears and cold compresses. To reduce contagiousness, povidone-iodine (PVP-I) 0.8% can be used (68). It has been demonstrated that PVP-I 0.6%/DEX 0.1% significantly improves clinical resolution in patients with acute adenoviral conjunctivitis (69).

Allergic conjunctivitis

Allergic conjunctivitis is a type I hypersensitivity reaction triggered by allergens like dust and pollen, causing histamine release, mast cell degranulation, and activation of proinflammatory mediators (70). Mast cell activation induces vascular endothelial changes, adhesion molecule expression, and secretion of chemokines such as MIP-1 alpha and IL-8, contributing to the late-phase reaction (71). Clinically, it presents with chemosis, lid crusting, bilateral watery discharge, and symptoms like itching, sneezing, cough, and nasal congestion. Blurry vision, photophobia, or pain are uncommon and warrant ophthalmology referral (72). Treatment targets acute and chronic inflammation (73). Conservative measures include cold compresses and antihistamine/vasoconstrictor drops, which reduce chemosis and vascular permeability (74). Mast cell stabilizers are effective in moderate cases, especially seasonal allergic conjunctivitis with corneal involvement (75). NSAIDs such as ketorolac decrease prostaglandin E2 in tears, reducing itching (76).

Neonatal conjunctivitis

Ophthalmia neonatorum (ON), also known as neonatal conjunctivitis, is an inflammation of the conjunctiva occurring within the first 4 weeks of life. ON is typically transmitted to the infant from the mother's birthing canal during delivery (77). Clinical signs include edema and erythema of the eyelids, purulent discharge, and inflammation of the palpebral conjunctivae (78). Many infectious cases are caused by sexually transmitted pathogens such as *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, but ON can also result from HSV-2, *Staphylococcus* species, *H. influenzae*, and other gram-negative organisms (77,79). Among these, conjunctivitis caused by *Neisseria gonorrhoeae* is particularly severe and, if left untreated, can rapidly progress to corneal ulceration and cause permanent vision loss (80). For prophylaxis, topical erythromycin, or an aminoglycoside like tobramycin, has largely replaced silver nitrate and is utilized within 24 hours of birth (81).

Treatment for gonococcal ON involves a one-time dose of ceftriaxone 125 mg intramuscular (IM) (82). For chlamydial ophthalmia or pneumonia, a two-week course of oral erythromycin is recommended (83).

Blepharitis

Blepharitis is a common inflammatory condition of the eyelids, characterized by a burning sensation, itching of the lid margins, flaking, and redness (72). Although rare, severe cases of blepharitis can lead to visual impairments secondary to keratopathy and corneal ulceration, as well as permanent alterations in eyelid morphology (84). Blepharitis is classified into two types based on location: anterior and posterior blepharitis (85). Anterior blepharitis affects the base of the eyelashes and their follicles, and is traditionally associated with staphylococcus, seborrheic dermatitis, and rosacea (86). Conversely, posterior blepharitis is primarily caused by underlying meibomian gland dysfunction (MGD), which is a condition marked by decreased glandular secretion and obstruction of the terminal ducts, reducing tear film stability and optical clarity (72,87). Marginal blepharitis refers to the combination of both anterior and posterior blepharitis, as there is frequently considerable overlap between the two types (88). Both posterior and anterior blepharitis can be caused by demodex mites, intracellular parasites that inhabit hair follicles and sebaceous glands (89). Maintaining proper eyelid hygiene is the cornerstone of treatment and is highly effective for managing blepharitis. Bacterial imbalance at the eyelid margin is believed to contribute to development of blepharitis, although the exact pathogenesis remains unclear. To date, bacitracin and erythromycin ointments are the most frequently used treatments (90).

Hordeolum

Hordeolum, or sty, is a purulent eyelid margin inflammation from sebaceous gland obstruction (91). External hordeolum affects the glands of Zeis, while internal hordeolum involves the meibomian glands (92). *Staphylococcus* species are the main cause and may occasionally cause cellulitis (93). Lesions often resolve spontaneously, but internal hordeola can become chronic or progress to chalazion if untreated (94). Standard care includes warm compresses and eyelid massage; topical antibiotics or mechanical drainage may be needed in refractory cases (95). Chronic conditions like diabetes or seborrheic dermatitis increase risk (96).

Endophthalmitis

Endophthalmitis is a severe form of ocular inflammation stemming from infection of the intraocular cavity and can lead to blindness if not treated promptly (97). The condition is classified as either “exogenous”; or “endogenous”; based on the source of the infection (98). Exogenous endophthalmitis arises when infecting agents penetrate the eye from outside the body, usually following penetrating trauma or intraocular surgery (99). Endogenous endophthalmitis originates from an internal source spreading through the bloodstream – this form is much less common (100). The condition usually presents suddenly with decreased visual acuity, pain, hypopyon, and eyelid edema (8). Most cases are caused by gram-positive bacteria (85.1%, mainly *S. epidermidis*), with gram-negative bacteria (10.3%) and fungi (*Candida*, *Aspergillus*, 4.6%) less common (101,102). Suspected cases require urgent ophthalmology referral. Diagnosis and treatment involve intraocular cultures, intravitreal antibiotics covering gram-positive and gram-negative bacteria, and sometimes systemic broad-spectrum antibiotics (103). Antifungals such as voriconazole, fluconazole, or amphotericin B are used when indicated (104). Severe cases may benefit from vitrectomy, which reduces infection and inflammation (105) and improves retinal oxygenation; vitrectomy is often more effective than intravitreal antibiotics alone in severe cases (106).

Orbital cellulitis (OC)

OC is a serious inflammatory condition, usually from an infectious source, affecting the contents of the globe and the surrounding soft tissues behind the orbital septum (107). While OC can occur at any age, it is more frequently observed in children (108). The most common causative microorganisms are *S. pneumoniae*, *S. aureus*, and *S. pyogenes* (109). Symptoms of orbital cellulitis include eyelid edema, conjunctival chemosis, proptosis, periorbital skin erythema, and decreased eye motility (110). The most common predisposing factor for orbital cellulitis across all age groups is secondary infection spreading from the paranasal sinuses (111). Other significant predisposing factors include intracranial infections, dental abscesses, and middle ear infections (112). On magnetic resonance imaging (MRI), orbital cellulitis is characterized by high signal intensity in T2-weighted fat-saturated sequences (113). Management of orbital cellulitis involves initiating intravenous broad-spectrum antibiotics and addressing any

associated sinusitis. Surgical drainage may be performed when the infection does not resolve with antibiotics alone. While this approach effectively controls active infection, inflammation sequelae may persist for weeks to months (114). If not adequately treated, orbital cellulitis can lead to serious complications such as intracranial abscesses, vision loss, and meningitis (115).

Uveitis

Uveitis is inflammation of the uvea, affecting the iris, ciliary body, or choroid, and is classified as anterior, intermediate, posterior, or panuveitis (116). Incidence does not differ significantly between sexes (117). Risk factors include infections (CMV, tuberculosis, toxoplasmosis, HSV, syphilis) and systemic diseases like spondyloarthropathies (118), with toxoplasma gondii a leading cause globally (119). Clinically, patients present with photophobia, pain, ciliary flush, and tenderness (120). Slit-lamp examination reveals inflammatory cells in the anterior chamber and protein-induced flare (121). Uveitis arises from an imbalance between regulatory and inflammatory immune mechanisms, influenced by genetic and environmental factors (122). Treatment depends on type: anterior uveitis responds to topical corticosteroids (123), with cycloplegics (e.g., homatropine) preventing synechiae and alleviating pain (124,125). Intermediate uveitis with macular edema may require periocular corticosteroids (126) or intravitreal implants like Ozurdex and Retisert (127). Table 2 provides a comparative overview of symptoms, causes, and treatments for major red eye conditions.

Eye injury

Hyphema

Hyphema is blood in the anterior chamber, often caused by trauma, coagulopathies, uveitis, juvenile conditions, or intraocular surgery (128). Complications include corneal blood staining, elevated intraocular pressure (IOP), optic nerve atrophy, and permanent vision loss; trauma can also cause angle recession glaucoma (129). Patients with conditions like sickle cell disease have a higher risk, though most recover fully (130). Treatment includes elevating the head to 30°, topical analgesics (tetracaine or proparacaine), rigid eye shielding, bed rest, and avoiding antiplatelet/anticoagulants (131). Daily follow-ups monitor vision, IOP, and slit-lamp findings due to rebleeding risk in the first

Table 2 Overview of red eye conditions

Items	Symptoms	Causes	Treatment
Scleritis	Intense pain and tenderness; localized or widespread redness; scleral thinning	Idiopathic; malignancy; surgically induced; medication side effects	Topical corticosteroid eye drops; oral NSAIDs; oral corticosteroids; subconjunctival corticosteroid injections; immunosuppressive medications
Episcleritis	No symptoms; localized or widespread redness	Idiopathic; autoimmune diseases (SLE, rheumatoid arthritis, Crohn disease, psoriatic arthritis)	Do not require continuous management; administer topical steroids; refer to a rheumatologist specializing
Keratitis	Sensitivity to light; decreased vision; redness, pain, and irritation in the eye	Bacterial (<i>Pseudomonas</i> , <i>Staphylococcus</i> , <i>Streptococcus</i>); viral (HSV, HZV); fungal (<i>Candida</i> , <i>Aspergillus</i> , <i>Fusarium</i>); protozoal (<i>Acanthamoeba</i>); helminth (<i>Onchocerca</i>)	Antibiotics (vancomycin and fluoroquinolones or tobramycin or gentamicin); PHMB (0.02%) and chlorhexidine (0.02%); topical 1% povidone-iodine, in combination with 0.1% dexamethasone; topical amphotericin-B
Conjunctivitis	Foreign body sensation, redness, eye irritation, photophobia, tearing; itchy, watery discharge	Bacterial (<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>S. aureus</i>); viral (adenovirus, enterovirus, HSV, VZV, EBV); allergic (pollens, dust, animal dander); neonatal (<i>C. trachomatis</i> , <i>N. gonorrhoeae</i>)	Antibiotics (ciprofloxacin, bacitracin, erythromycin); antivirals (trifluridine, ganciclovir); cold compresses; artificial tears
Uveitis	Red, painful eye; increasing floaters; decreased vision; chorioretinal lesions	Idiopathic 50%; ankylosing spondylitis, IBD, Reiter's syndrome, psoriatic arthritis; herpes virus	Determine the potential cause; administer topical steroids; use systemic immunosuppressive medications (cyclophosphamide, azathioprine, methotrexate); apply immunomodulating agents
Blepharitis	Itching, burning, and crusting of the eyelids; tearing, blurred vision, and a sensation of a foreign body in the eye	Bacterial (<i>Staphylococcus</i>); viral (herpes simplex, varicella zoster)	Topical antibiotics; good hygiene practices; warm compresses
Endophthalmitis	Poor visual acuity; difficulty visualizing the fundus; vitreous inflammation; corneal edema	Bacterial (<i>S. epidermidis</i> , <i>S. viridans</i> , <i>S. aureus</i>); fungi	Collect a vitreous sample for culture; inject intravitreal antibiotics; apply topical antibiotics; use vancomycin and ceftazidime; perform surgery
Hordeolum	Burning, tender swelling on the eyelid; lid edema and erythema	<i>S. aureus</i> ; <i>S. epidermidis</i>	Self-limiting condition; warm compresses; erythromycin
Orbital cellulitis	Periorbital redness, swelling, and eyelid edema	<i>S. aureus</i> ; <i>S. pneumoniae</i> ; <i>S. pyogenes</i> ; <i>H. influenzae</i>	Administer intravenous antibiotics; conduct blood cultures; perform surgery
Hyphema	Eye pain, nausea, vomiting; intraocular foreign object; vision changes; photophobia	Trauma; von Willebrand disease; hemophilia;	Bed rest to prevent re-bleeding; apply topical steroids to reduce inflammation; use antifibrinolysis agents; perform surgical evacuation if the intraocular pressure increases

Table 2 (continued)

Table 2 (continued)

Items	Symptoms	Causes	Treatment
Pterygium	Cosmetic blemish; reduced vision; sensation of a foreign object in the eye	UV rays from sunlight; eye irritation due to dry and dusty conditions	Lubricate with artificial tear drops; apply topical NSAIDs; perform surgery
Subconjunctival hemorrhage	Normal; may experience photophobia, foreign body sensation, and headache	Trauma; hypertension; ocular surgeries; local anesthesia	Artificial tears; ice packs; brimonidine; oxymetazoline
Trichiasis	Sensation of a foreign body; pain; eye irritation and redness	Misdirection of the eyelash follicle, causing the lash to grow inward towards the cornea	Electrolysis
Chemical injuries	Mild irritation; eye redness and pain; blindness	Often occurs at home, typically from drain cleaners, detergents, disinfectants, solvents, and cosmetics	Rinse immediately with clean water; inspect and remove any foreign objects; apply lubricants frequently; take oral pain medication
Dry eye	Excessive blinking; sensation of pressure and stinging in the eyes; gritty or foreign body sensation	Decreased muco-aqueous component; environmental factors; systemic illnesses	Lubricating eye drops; artificial tears; dietary modifications

EBV, Epstein-Barr virus; HSV, herpes simplex virus; HZV, herpes zoster virus; IBD, inflammatory bowel disease; NSAIDs, nonsteroidal anti-inflammatory drug; PHMB, polyhexamethylene biguanide; SLE, systemic lupus erythematosus; VZV, varicella zoster virus.

10 days (8). Surgery is indicated for persistent high-grade hyphema, uncontrolled glaucoma, or elevated IOP (132).

Corneal abrasion

Corneal abrasion, also known as corneal epithelial defect, occurs when the top layer of the cornea, the corneal epithelium, is damaged or lost, typically due to non-penetrating trauma (133). Symptoms include reduced visual acuity, redness, excessive tearing, and photophobia, making corneal abrasions a common occurrence in emergency department settings (134,135). In addition to assessing for foreign bodies, a slit-lamp examination with fluorescein dye aids in the diagnosis and visualization of the abrasion's location and size (8). Healing involves a complex process of cell migration, death, proliferation, differentiation, and extracellular matrix remodeling. Healing the corneal endothelium is particularly difficult as these cells do not regenerate. Rather, they rely on cell migration, spreading, and possible epithelial-mesenchymal transformation (136). Treatment typically involves aggressive lubrication, antibiotic drops or ointments with broad-spectrum coverage, and daily follow-up until recovery (133). Commonly used antibiotics include ofloxacin, ideal for its *P. aeruginosa* coverage in contact lens wearers, erythromycin,

and polysporin (133,137). Although research regarding pain management continues to evolve, oral analgesics and a short course of dilute topical anesthetics such as tetracaine and proparacaine within the first 24 hours carries minimal risk and can minimize pain in patients (138,139). However, because not all ophthalmologists support routine outpatient use, this method is typically saved for minor, straightforward abrasions and is still debatable.

Subconjunctival hemorrhage (SCH)

SCH is a common cause of acute, painless red eye frequently seen in ophthalmology clinics (140). It occurs when blood from conjunctival or episcleral vessels leaks into the subconjunctival space (141). SCH can result from systemic diseases, trauma, or the rupture of fragile vessels after sudden pressure increases, such as vomiting or Valsalva maneuvers (142,143). Elderly patients with vascular conditions like arteriosclerosis, hypertension, and diabetes often have weakened conjunctival vessels prone to rupture. While trauma typically causes localized bleeding at the injury site, often in the temporal region, SCH tends to appear in the inferior and temporal areas of the conjunctiva (141). Treatment is generally not required, as the condition resolves on its own within 1–2 weeks, though recovery may

extend up to 4 weeks in patients on anticoagulants (144).

Trichiasis

Trichiasis is a condition in which eyelashes grow abnormally towards the eye's surface, causing discomfort. It can present as a few misdirected lashes in either a segmental or diffuse pattern (145). While trichiasis can occur at any age, it is more often seen in adults and presents with symptoms of pain, foreign body sensation, redness, and tearing in the affected eye (146). Major causes of trichiasis include chronic eyelid margin inflammation, skin disease, conjunctival disease, and scarring from surgery or trauma (147). If left untreated, trichiasis can lead to corneal ulceration, scarring, and ultimately, vision impairment (148). Treatment options vary from simple measures like epilating the misdirected lashes or using a bandage contact lens to more permanent solutions such as destroying the problematic lash roots with argon laser or electrolysis (149).

Pterygium

Pterygium is a benign fibrovascular growth extending toward the cornea, causing astigmatism, vision impairment, and recurrent inflammation (150). Risk factors include ultraviolet (UV) exposure, male sex, and age (151). It usually arises nasally in the interpalpebral zone, presenting with redness, dryness, irritation, and decreased vision, and is unique to humans (152). Vascular endothelial growth factor (VEGF) receptor 2 expression may predict recurrence (153), while matrix metalloproteinases (MMPs) and tissue inhibitors of MMPs (TIMPs) at the leading edge contribute to inflammation and corneal invasion (154). Redness and edema can be reduced with vasoconstrictors, and antihistamines prevent pruritus (155). Subconjunctival 5-fluorouracil halts progression in recurrent cases (156). Surgery is indicated if the visual axis is affected, astigmatism is significant, symptoms persist, or for cosmetic reasons (8).

Chemical injuries

Ocular chemical burns are an emergency that can severely impair vision and quality of life (157). Severity depends on exposure duration and chemical type (158). Tissue damage persists while chemicals remain in contact with the eye (159). Household disinfectants and cleaning agents are common pediatric causes (159). Alkalis cause more severe injury than acids, affecting both external and internal structures

(160,161). Immediate treatment with thorough irrigation using clean water or isotonic solutions until surface pH reaches ~7 is critical (162). Irrigation should last at least 10 minutes, with a Morgan Lens allowing hands-free prolonged irrigation (163). Slit-lamp examination with fluorescein assesses corneal and conjunctival damage. Severe cases may require platelet-rich plasma drops or topical biologics to restore function (164).

Other

Glaucoma

Glaucoma is a progressive loss of retinal ganglion cells with optic nerve changes (165). It results from impaired aqueous humor outflow due to trabecular meshwork resistance or angle occlusion, causing elevated IOP (166,167). As the second leading cause of irreversible blindness in the USA, it is a major public health concern (168). Primary angle-closure glaucoma (PACG) occurs when the peripheral iris blocks aqueous flow, potentially progressing to AACG with sudden vision loss, severe eye pain, and headaches (169). IOP ≥ 30 mmHg warrants suspicion for AACG; IOP > 40 mmHg with acute symptoms requires urgent ophthalmology referral. Workup includes visual fields, pupillary testing, slit-lamp exam, tonometry, gonioscopy, and optic nerve assessment (8). Treatment aims to lower IOP (170). Open-angle glaucoma is managed with medications like carbonic anhydrase inhibitors or trabeculoplasty (171,172). In PACG/AACG, urgent initiation of topical glaucoma medications (i.e., beta blockers, alpha-2 agonists, cholinergic agonists, and prostaglandin analogues), systemic carbonic anhydrase inhibitors, and mannitol is indicated. Definitive laser peripheral iridotomy relieves pupillary block, and severe or refractory cases may require drainage devices or trabeculectomy (8). Prophylactic laser peripheral iridotomy (LPI) is performed in the contralateral eye due to increased risk (173).

Dry eye syndrome (DES)

DES is a multifactorial disorder affecting tears and the ocular surface, impairing comfort and vision (174). The tear film has three layers: lipid (prevents evaporation), aqueous (moisturizes), and mucin (ensures even distribution) (175). It nourishes, lubricates, and removes debris from the eyes (176). Symptoms include blurred vision, burning, and pain, which can affect quality of life (177). If left untreated, DES can lead to persistent visual disturbance, discomfort, and potential

damage to the eye's surface. Both advancing age and female gender have been identified as significant risk factors for developing DES, though the exact pathogenesis remains unclear (178). Additionally, DES is often associated with systemic diseases, further complicating its management (179). Diagnosis involves tear-secretion tests, tear meniscus evaluation, and tear film break-up time (TBUT <5 sec indicates short TBUT-type DES) (180). Fluorescein staining may reveal punctate epithelial erosions in severe cases (181). Management includes tear retention drops, artificial tears, and gels (182).

Strengths and limitations

Strengths

This review provides a comprehensive, up-to-date synthesis of red eye presentations specifically aimed at primary care and frontline clinicians. By integrating recent advances in diagnosis and management with practical clinical frameworks, the article offers a valuable resource to improve the timely identification and treatment of serious ocular conditions. Unlike previous reviews, this article consolidates findings across a broad range of etiologies while maintaining an accessible, structured approach designed for non-ophthalmologists.

Limitations

As a review, this article may be subject to selection bias in the choice of referenced studies. While we aimed to include the most relevant and recent evidence, a systematic review methodology was not employed. Additionally, rapid advances in ophthalmic diagnostics and therapeutics mean that recommendations may require future updates as new technologies and treatment protocols emerge. Understanding the strengths and limitations of this review provides important context for applying its clinical recommendations and highlights areas for future investigation.

Use of topical steroids in ophthalmic emergencies

Although ocular inflammatory diseases such as scleritis, moderate to severe episcleritis, and uveitis can be effectively managed with topical corticosteroids, care must be taken when using them in emergency or primary care settings. An undiagnosed infectious etiology, such as bacterial, fungal, or herpetic keratitis, may exacerbate with steroid

use, causing irreversible vision loss or delayed epithelial healing if the medication is administered improperly or too soon. Hence, unless there is a well-established follow-up and a clear, documented diagnosis, primary care physicians should refrain from starting topical steroids without first performing an ophthalmologic evaluation. The advantages and disadvantages of using steroids should be carefully considered when the inflammation is severe and threatening to the patient's vision and an ophthalmology consultation is not possible right away. Steroids should, whenever feasible, be postponed until after corneal scraping and microbiological analysis, particularly in cases of suspected infectious keratitis.

Conclusions

Red eye is a common chief concern in both primary care and emergency settings, representing a broad spectrum of potential etiologies. It is important for healthcare providers to quickly and accurately distinguish benign causes from sight-threatening ones early on to initiate proper management. For example, allergic conjunctivitis is often self-limiting, while BK or uveitis requires prompt attention and targeted treatment. Differentiation can be best accomplished through thorough history-taking, detailed physical examination, and relevant diagnostic testing. Workup to rule out underlying systemic conditions, such as autoimmune diseases linked to scleritis, episcleritis, or uveitis, should also be considered when assessing for causes of red eye and the need for multidisciplinary involvement. Early detection and diagnosis can guide treatment options and mitigate the risk of complications. A basic examination of the pupils and their constriction response can effectively identify those needing urgent ophthalmologic referral. Immediate referral is also necessary for any patients experiencing visual blurring and significant eye pain, despite these symptoms not being systematically studied in all contexts. Management varies significantly based on underlying pathology but often includes antibiotics, steroids, anti-inflammatory drugs, anti-virals, antifungal agents, and supportive care such as lubricants and compresses. In addition to an individualized management approach, patient education plays a crucial role in further prevention of red eye and its course. To reduce disease burden, healthcare providers should encourage awareness of alarming symptoms while emphasizing regular eye examinations, hygiene, and proper contact lens usage. In this review, we highlighted effective examination and

management strategies for 22 common causes of red eye, emphasizing practical clinical approaches that support accurate diagnosis, timely intervention, and appropriate referral. By providing a structured framework, this review aims to enhance clinician confidence, reduce misdiagnosis, and improve patient outcomes in the management of red eye presentations across diverse clinical environments.

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